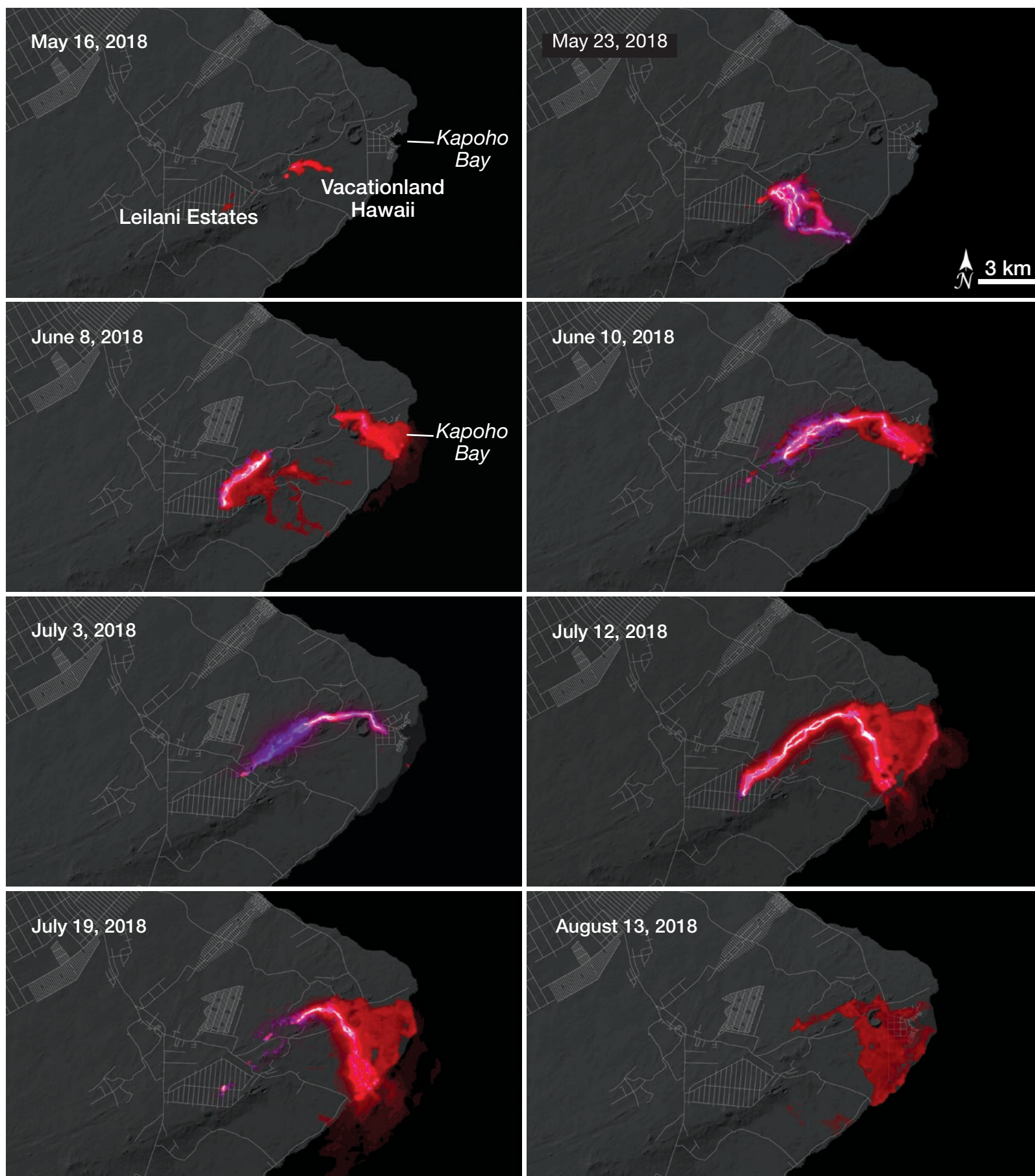






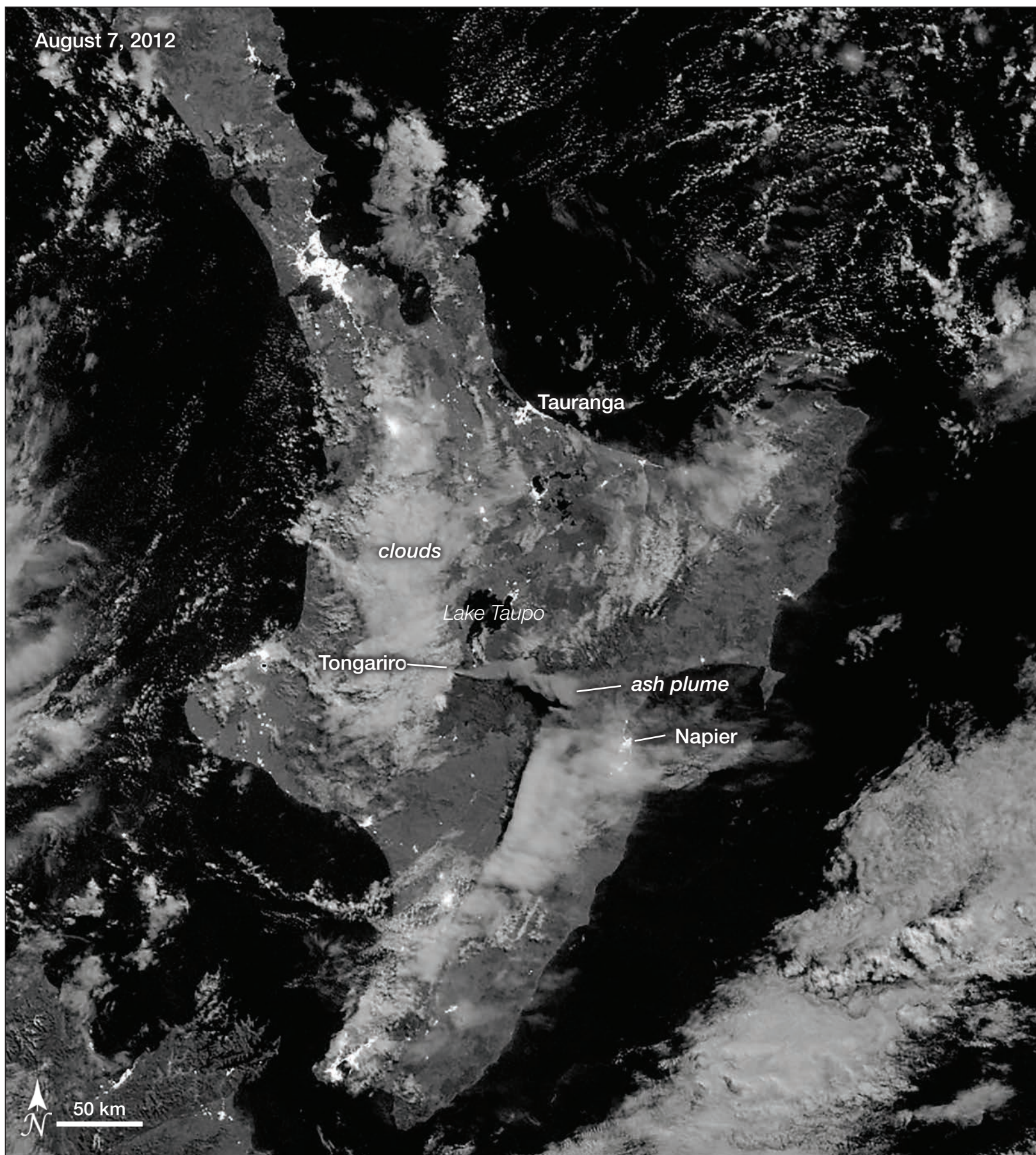
Mount Tongariro Erupts—New Zealand

Late on August 6, 2012, New Zealand's Mount Tongariro erupted for the first time in 115 years, spewing a cloud of ash over North Island, closing roads, and cancelling domestic flights. The image at right, observed by the VIIRS DNB on Suomi NPP, shows Mount Tongariro as it appeared at 12:55 a.m. New Zealand time on August 7.

The volcano lies in a mostly undeveloped part of the country. The 6,490-foot (1,978-meter) peak is a popular spot for hikers and is right next to Mount Ngauruhoe, the stand-in for Mount Doom in the "Lord of the Rings" movies. Both Tongariro and Ngauruhoe are part of the circum-Pacific seismic belt or "Ring of Fire" region, known for its seismic activity and volcanism.

Tongariro sent ash at least 20,000 feet (6,100 meters) into the air. Because of the impact to aircraft operations, the New Zealand Civil Aviation Authority issued advisories to reroute air traffic around the ash plume so as to avoid damage to aircraft engines as happened in 1989 when a KLM Boeing 747 commercial jet flew into an ash plume produced by Alaska's Mt. Redoubt, which resulted in damage to all four engines. Swift, skillful piloting kept this incident from being a tragedy, as all engines were subsequently restarted, and the aircraft landed safely.

The first appearance of the ash plume occurred far downwind of Tongariro in thermal imagery, as ash was lofted high enough to produce a strong thermal contrast in the atmosphere. Images from the VIIRS DNB can be combined with thermal images such as from VIIRS (not shown here) to give a more-complete view of such activity on Earth's surface. Getting thermal data closer to a volcano with conventional infrared sensors is difficult, but VIIRS can detect such signals close to the source and lower in the atmosphere than other sensors.





Nighttime Glow at Mount Etna—Italy

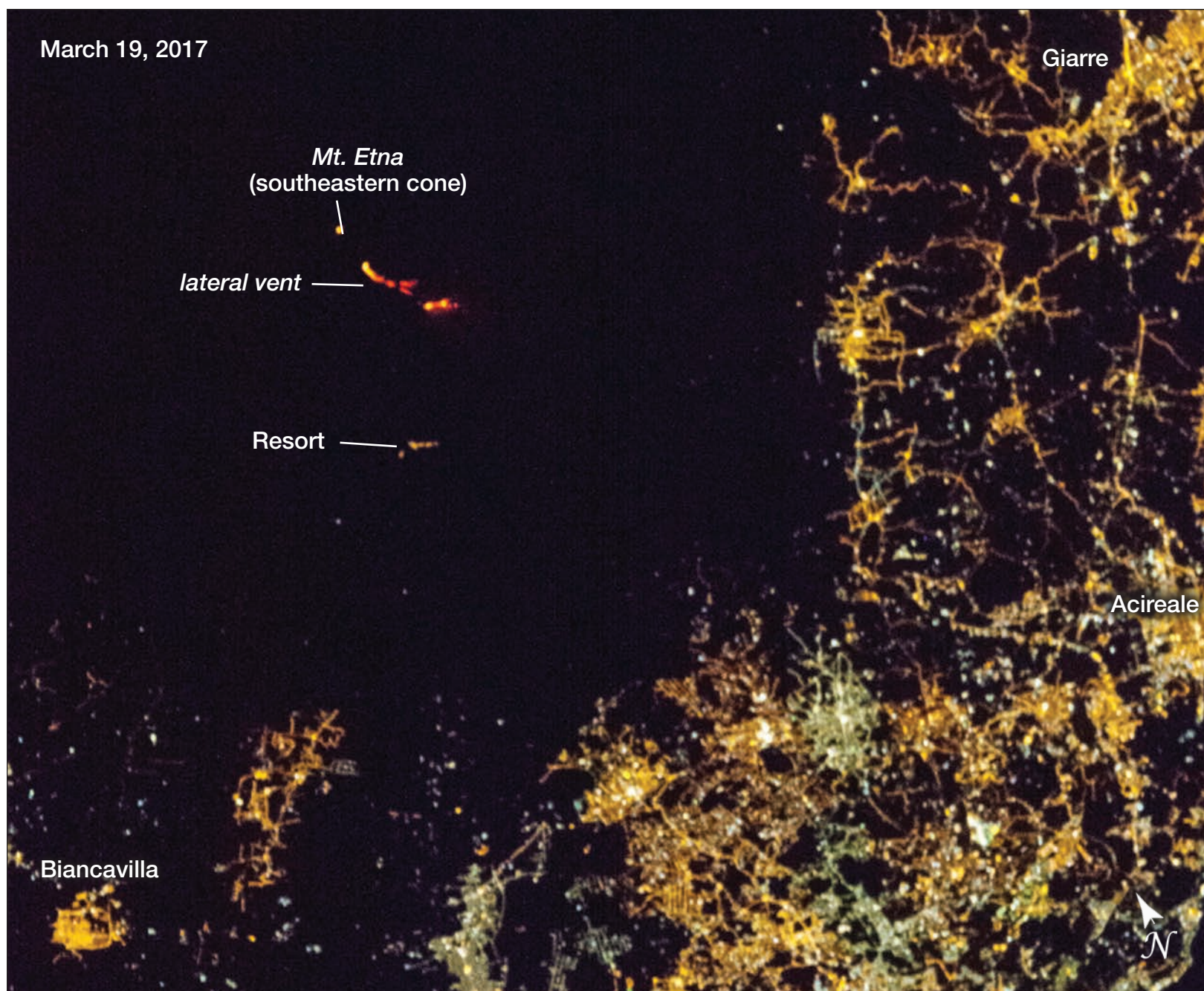
At about 2:30 a.m. local time on March 16, 2017, the VIIRS DNB on the Suomi NPP satellite captured this nighttime image of lava flowing on Mount Etna in Sicily, Italy. Etna is one of the world's most active volcanoes.



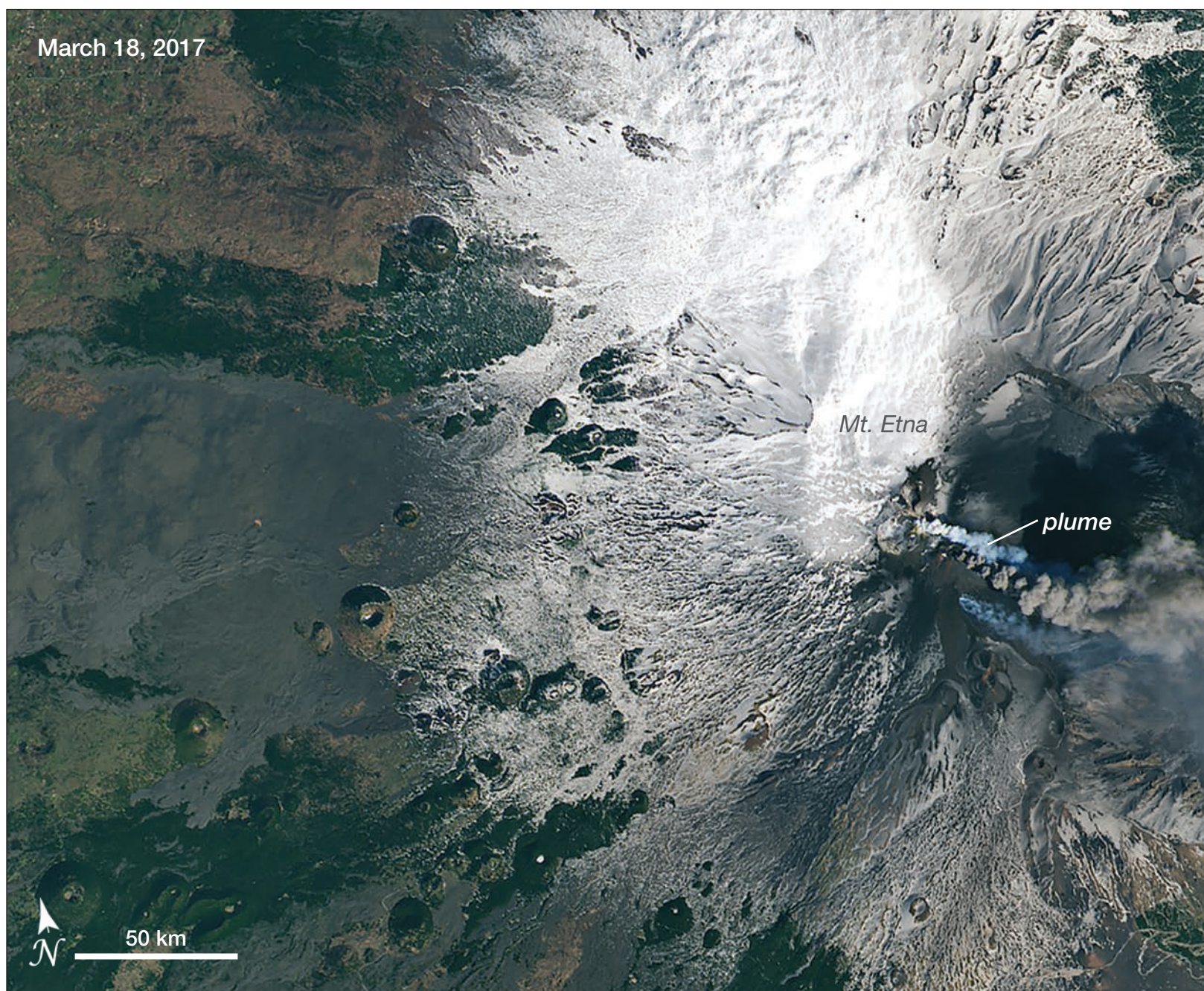


Mount Etna Erupts—Italy

The highly active Mount Etna in Italy sent red lava rolling down its flank on March 19, 2017. An astronaut onboard the ISS took the photograph below of the volcano and its environs that night. City lights surround the mostly dark volcanic



mountain. On the previous day, the Operational Land Imager (OLI) on Landsat 8 acquired this natural-color image (below) of an ash plume emanating from the summit and the flank.

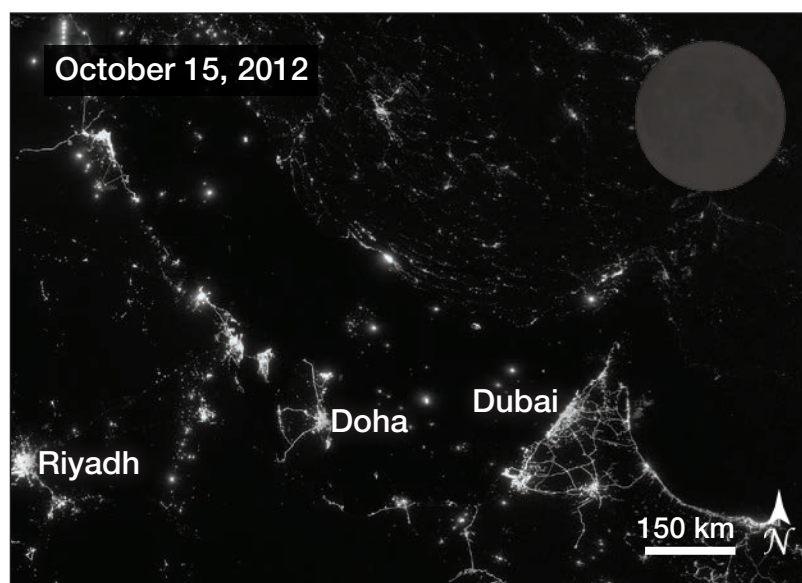
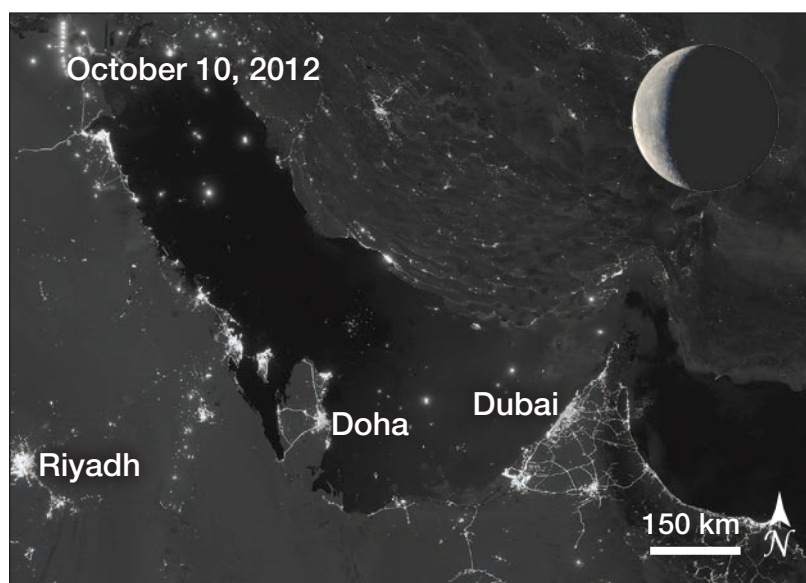
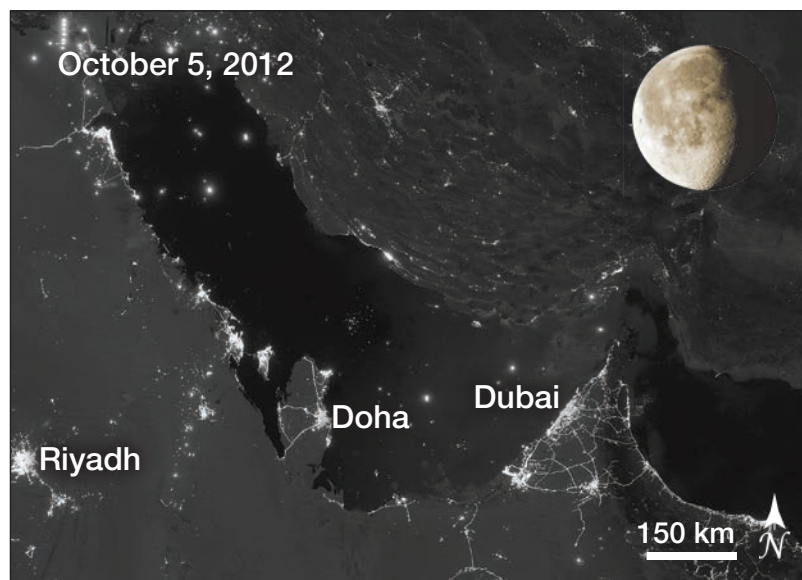
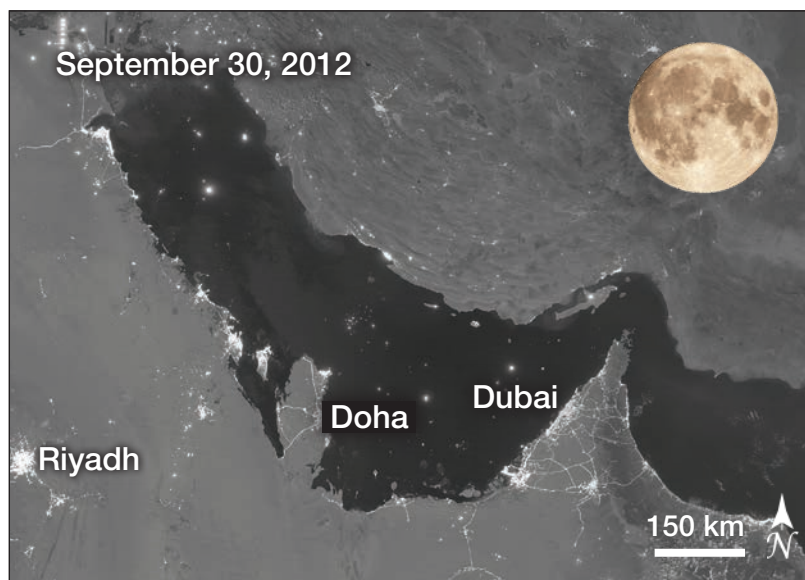




Moonlight and Moonglint

Moon Phases—Persian Gulf

The VIIRS DNB on the Suomi NPP satellite captured these nighttime views of the Persian Gulf region on September 30 and October 5, 10, and 15, 2012. Each image includes an inset of the Moon in one of four different phases. September 30 shows the Persian Gulf by the light of the full Moon; October 15 shows the effects of a new Moon. As the amount of moonlight decreases, some land surface features become harder to detect, but the lights from cities and ships become more obvious. Urbanization is most apparent along the northeastern coast of Saudi Arabia, in Qatar, and in the United Arab Emirates (UAE). In Qatar and UAE, major highways can even be discerned by nighttime lights.





Moonglint—Elba and the Mediterranean

A crew member onboard the ISS took this photograph of the northern Mediterranean Sea and some coastal Italian towns and islands on October 17, 2013. The reflection of the Moon on the sea surface—moonglint—reveals highly complex patterns. The strongest reflection is near the center of the Moon’s disc, which brightens the water around the island of Elba. In these complex patterns, the dark areas of the sea surface can sometimes make islands, such as Montecristo and Pianosa, harder to see.

The reflection off sea surfaces captures many different natural processes but also some made by humans. North of Elba, waves trailing behind ships make a classic V-shaped pattern. The meandering line coming off Montecristo Island is an island wake, a result of alternating vortices of wind that develop on the downwind side of the island. This wake is the strongest evidence that a northeast wind was blowing (right to left in this image) on the night of the photo. A shorter, meandering wind pattern is being shed off Punta Ala on the mainland. Smoother surfaces, protected from wind, are usually brighter because they are a better mirror for moonlight.

The sea surface also displays numerous tight swirls known as gyres. The broad swath of parallel lines (top left) is probably part of the larger circulation of the sea, which usually experiences north-flowing currents around Elba.